DOHYUN HWANG (黄棹炫)

Undergraduate Researcher @ Yonsei University Mirae Campus

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I am an undergraduate researcher focused on artificial intelligence, federated learning, autonomous driving, topological data analysis, and medical data analysis. I approach research through technically grounded experimentation, cross-domain applicability, and careful translation from methodological ideas to realistic use cases.

EXPERIENCE

2024 - Present

CTO / AI & Web Developer

Futurevle.Co

- Led AI and web development initiatives across company projects.
- Contributed to technical decision-making, system design, and implementation planning.
- Worked on AI-powered services and web-based development projects with a focus on practical deployment.

EDUCATION

2022 - Present

Yonsei University Mirae Campus

Undergraduate Student

Pursuing rigorous training in computing, mathematics, and applied AI. Current coursework explores algorithms, system programming, AI mathematics, and data structures.

Mar 2019 - Feb 2022

Hyundai High School

- Developed strong interests in science, mathematics, computing, and analytical thinking.
- Received school-level awards and recognitions through academic and extracurricular activities.
- Grade 10: Korean Language, Mathematics, English, Korean History, Integrated Social Studies, Integrated Science, Scientific Inquiry and Experimentation, Informatics, Japanese I, Physical Education, Music, Art Creation
- Grade 11: Literature, Reading, Mathematics I, Mathematics II, Probability and Statistics, English I, English II, Physics I, Chemistry I, Life Science I, Exercise and Health, Music
- Grade 11 Career Elective: Geometry, Life and Science, Japanese II
- Grade 12: Speech and Composition, Calculus, English Reading and Writing, Korean History, Life and Ethics, Education, Essay Writing, Sports Life
- Grade 12 Career Elective: Classic Reading, Mathematical Inquiry and Research, Social Issues Exploration, Life Science II, Advanced Life Science

PUBLICATIONS

Accepted

Adversarial Robustness Analysis of Deep Learning-Based Automatic Modulation Classification in Wireless Communication

IEEE ICAIIC 2026

Sunjun Hwang, Eunho Choi, and Dohyun Hwang. Accepted Dec 21, 2025.

Contributed to the study of adversarial robustness in DL-based automatic modulation classification systems for wireless environments.

CURRENT RESEARCH

2025 - Present

Autonomous Driving + Federated Learning (Heterogeneous Models)

- Investigating federated learning settings in which heterogeneous models are trained within the same driving environment.
- Analyzing knowledge transfer dynamics, representation compatibility, and performance convergence across different model families.

2026 - Present **Topological Data Analysis for Medical Data Analysis**

- Applying TDA to BRCA breast cancer datasets to identify latent geometric and relational structures not readily captured by conventional statistics.

2026 - Present **Mamba-Based Autonomous Vehicle**

- Exploring Mamba-based State Space Model architectures for perception and decision-making in autonomous driving.

AWARDS & HONORS

2026 **HIRA Big Data Global Talent Program, Grand Prize**
Health Insurance Review & Assessment Service | HIRA President's Award

2025 **Gangwon Province Startup Competition, Grand Prize**
Gangwon Special Self-Governing Province | Governor's Award

2025 **Yonsei MEDICI+ Startup Hackathon, Grand Prize**
Yonsei University